C 20650 **(Pages : 2)** **Name.....**

Reg. No.....

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS–UG)

Mathematics

MTS 6B 14 (E01)—GRAPH THEORY

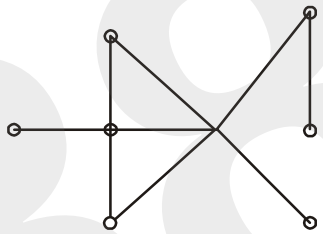
(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

Section A (Short Answer Type Questions)

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. Find number of edges of $k_{m,\,n}$.
2. Draw the graph $K_6 - \{v\}$ where v is any vertex in K_6.
3. Draw a 4-regular graph with ten vertices.
4. Define intersection of two graphs.
5. Let G be a simple graph in which there is no pair of adjacent edges. What can you say about the degree of the vertices in G ? Justify.
6. Draw the graph $K_{2,3,3}$.
7. Define eccentricity and radius.
8. Draw Peterson graph and find a trial of length 5.
9. When can you say that the complete graph $kn, n \geq 3$ is Euler ? Justify.
10. Prove that any subgraph of a planar graph is planar.
11. Find K (G) for the graph.



12. How many different Hamiltonian cycles does K_n have ?

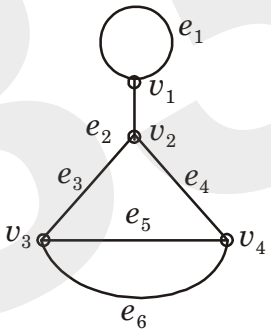
(8 × 3 = 24 marks)

Turn over

Section B (Paragraph/Problem Type Questions)

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. If G is a simple planar graph then prove that G has a vertex v of degree less than 6.
14. Prove that the complete bipartite graph $k_{3,3}$ is non-planar.
15. Let G be a graph with n vertices, where $n \geq 2$. Then prove that G has atleast two vertices which are not cut vertices.
16. Explain the Konigsberg bridge problem.
17. Prove that G is connected if and only if it has a spanning tree.
18. Let G be a graph with n vertices. Then prove that G is a tree if and only if G is a connected graph with $n - 1$ edges.
19. Define (i) adjacency matrix of a graph G ; (ii) incidence matrix of a graph G. Find the adjacency and incidence matrix of the following graph G.



(5 × 5 = 25 marks)

Section C (Essay Type Questions)

Answer any **one** *question.*
The question carries 11 *marks.*

20. Prove the following :
    (i) A connected graph G has an Euler tail if and only if it is atmost two odd vertices.
    (ii) A simple graph G is Hamiltonian if and only if it closure c (G) is Hamiltonian.
21. Prove that, a non-empty graph with atleast two vertices is bipartite if and only if it has no odd cycle.

(1 × 11 = 11 marks)